Probing Non-Gaussian Cosmic Fields with Critical-Point Statistics

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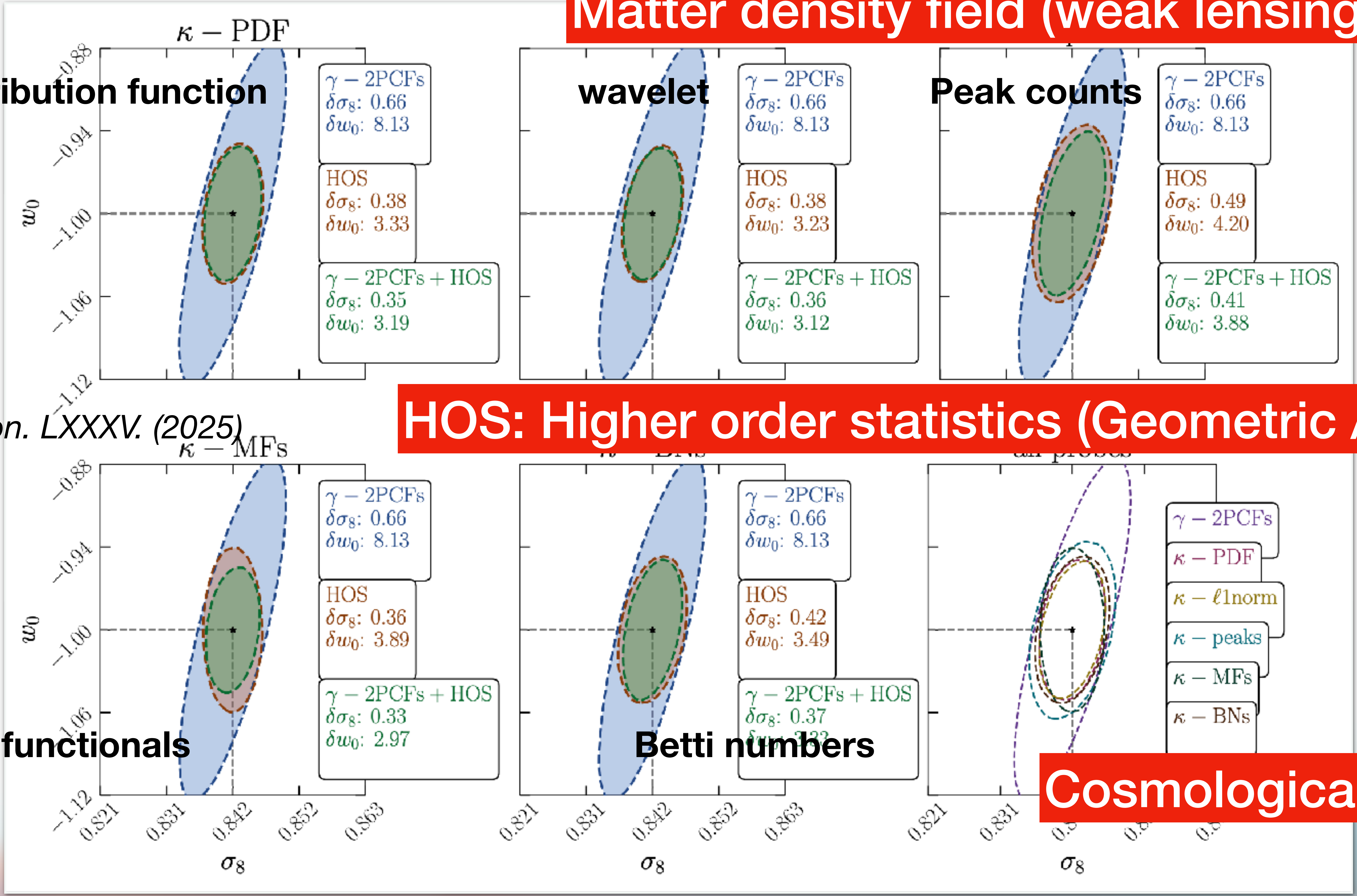
Based in part on Y. He and T. Matsubara,
"Validating weakly non-Gaussian critical-point statistics in cosmic fields",
submitted to Physical Review D (2026).

Schematic illustration generated with ChatGPT

Motivation: Information beyond two-point statistics

Matter density field (weak lensing, LSS)



Probability distribution function

wavelet

Peak counts

Euclid Preparation. LXXXV. (2025)

HOS: Higher order statistics (Geometric / Topological)

Minkowski functionals

Betti numbers

Cosmological information

Critical points classification in 3D density fields

- Normalize a density field $f(\boldsymbol{x})$ as

$$\alpha(\boldsymbol{x}) \equiv \frac{f(\boldsymbol{x})}{\sigma_0}, \qquad \boldsymbol{\eta}(\boldsymbol{x}) \equiv \frac{\nabla f(\boldsymbol{x})}{\sigma_1}, \qquad \zeta_{ij} \equiv \frac{\partial_i \partial_j f(\boldsymbol{x})}{\sigma_2} \text{ (Hessian matrix)}$$

$$\sigma_0^2 = \langle f^2 \rangle$$
$$\sigma_1^2 = \langle (\nabla f)^2 \rangle$$
$$\sigma_2^2 = \langle (\nabla\nabla f)^2 \rangle$$

- $\boldsymbol{\eta} = 0 \quad \Rightarrow \quad$ critical points classified by ζ eigenvalues' sign
  - 2D: peak - -, saddle -+, void ++
  - ***3D: peak - - -, filament - - +, wall -++, void +++***

number density

analytic formula ?

Main contributions and outline

- ***Critical-point statistics***

★: different critical point

  - Extend kernels of leading-order peak number density to all critical points $\langle n^{\star}(\nu)\rangle$
    - Based on Gaussian: *BBKS (1986)*; non-Gaussian: *T. Matsubara (2020)*
- ***Numerical implementation***
  - Verify with N-body simulation fields using direct finding algorithm
    - Leading-order correction accurate regime
- ***Cosmological parameters inference***
  - Parameter sensitivity, posterior constrain in viable regime

Critical-point statistics: isotropic Gaussian fields

- Peak number density above a threshold $(\alpha > \nu)$:

$$\boldsymbol{Y} = \left(\alpha \equiv f/\sigma_0,\ \boldsymbol{\eta} \equiv \nabla f/\sigma_1,\ \zeta_{ij} \equiv \partial_i\partial_j f/\sigma_2\right)$$

$$\underbrace{\bar{n}^{\rm pk} \equiv \langle n^{\rm pk}(\nu)\rangle}_{\text{normalized by } \sigma_0} = \underbrace{\left(\frac{\sigma_2}{\sigma_1}\right)^3}_{\text{normalization factor}} \langle \Theta(\alpha - \nu)\, \underbrace{\delta_{\rm D}^3(\boldsymbol{\eta})}_{\boldsymbol{\eta} = 0}\, \Theta(\lambda_3) \underbrace{\left|\det \zeta\right|}_{\text{Jacobian}} \rangle$$

BBKS (1986)

Θ: step function, $-\zeta$ eigenvalues: $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$

- Joint Gaussian probability distribution:

$$\mathscr{P}_{\rm G}(\boldsymbol{Y}) \propto \exp\left[-\frac{\alpha^2 + J_1^2 - 2\gamma\alpha J_1}{2(1-\gamma^2)} - \frac{3\eta^2}{2} - \frac{5J_2}{2}\right], \text{ with } \gamma = \sigma_1^2/(\sigma_0\sigma_2)$$

- Rotational invariants: $\boldsymbol{Y} = \left(\alpha,\ \eta^2 \equiv \boldsymbol{\eta}\cdot\boldsymbol{\eta},\ J_1 \equiv -\zeta_{ii},\ J_2 \equiv \frac{3}{2}\underbrace{\tilde{\zeta}_{ij}}_{\zeta_{ij} - \delta_{ij}\zeta_{kk}/3}\tilde{\zeta}_{ji}\right)$

Gay et.al (2012)

Critical-point statistics: weakly non-Gaussian fields

T. Matsubara (2020)

- Edgeworth expansion: $\mathscr{P}(x) = \left(1 + \frac{\sigma_0}{6} S_3 H_3\right)\mathscr{P}_{\mathrm{G}}(x) + \mathscr{O}(\sigma_0^2)$, where $S_3 = \frac{\langle f^3 \rangle}{\sigma_0^4}$, $H_3 = -\frac{\sigma_0^3}{\mathscr{P}_{\mathrm{G}}}\frac{d^3\mathscr{P}_{\mathrm{G}}}{dx^3}$

- Wierner-Hermite expansion: $\mathscr{P}[f] = \mathscr{P}_{\mathrm{G}}[f] + \frac{1}{6}\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3} \langle \tilde{f}_{\boldsymbol{k}_1}\tilde{f}_{\boldsymbol{k}_2}\tilde{f}_{\boldsymbol{k}_3}\rangle_{\mathrm{c}} \mathscr{H}_3(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3)\mathscr{P}_{\mathrm{G}}$

$$\frac{-1}{\mathscr{P}_{\mathrm{G}}}\frac{\delta^n \mathscr{P}_{\mathrm{G}}}{\delta\tilde{f}_{\boldsymbol{k}_1}\delta\tilde{f}_{\boldsymbol{k}_2}\delta\tilde{f}_{\boldsymbol{k}_3}}$$

- Leading order: $\langle n^\star[f]\rangle = \langle n^\star\rangle_{\mathrm{G}} + \frac{1}{6}\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3} \langle \tilde{f}_{\boldsymbol{k}_1}\tilde{f}_{\boldsymbol{k}_2}\tilde{f}_{\boldsymbol{k}_3}\rangle_{\mathrm{c}}\, \mathscr{G}_3^\star(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3)$

$$\mathscr{G}_3^\star \equiv \left\langle \frac{\delta^3 n^\star}{\delta\tilde{f}_{\boldsymbol{k}_1}\delta\tilde{f}_{\boldsymbol{k}_2}\delta\tilde{f}_{\boldsymbol{k}_3}}\right\rangle_{\mathrm{G}} = -\int d\boldsymbol{Y}\, n^\star(\boldsymbol{Y})\hat{\mathscr{D}}(\boldsymbol{k}_1)\hat{\mathscr{D}}(\boldsymbol{k}_2)\hat{\mathscr{D}}(\boldsymbol{k}_3)\mathscr{P}_{\mathrm{G}}(\boldsymbol{Y}),$$

$$\frac{\delta}{\delta\tilde{f}(\boldsymbol{k})} \to \frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{ik_i}{\sigma_1}\frac{\partial}{\partial\eta_i} - \frac{k_ik_j}{\sigma_2}\frac{\partial}{\partial\zeta_{ij}} \equiv \hat{\mathscr{D}}(\boldsymbol{k})$$

  - Skewness parameters: $\langle \tilde{f}_{\boldsymbol{k}_1}\tilde{f}_{\boldsymbol{k}_2}\tilde{f}_{\boldsymbol{k}_3}\rangle_{\mathrm{c}} \times \boldsymbol{k}$ mode couplings

    Ex: $\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3} \langle \tilde{f}_{\boldsymbol{k}_1}\tilde{f}_{\boldsymbol{k}_2}\tilde{f}_{\boldsymbol{k}_3}\rangle_{\mathrm{c}}(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)k_3^2 = \langle(\nabla f\cdot\nabla f)\Delta f\rangle_{\mathrm{c}} \equiv S^{(2)}$

$\boldsymbol{Y} = (\alpha, \eta^2, J_1, J_2)$

$$\boxed{\langle n^\star(\nu)\rangle \Rightarrow \sum S\int d\boldsymbol{Y}\, n^\star\hat{\mathscr{D}}\hat{\mathscr{D}}\hat{\mathscr{D}}\mathscr{P}_{\mathrm{G}}}$$

Critical-point statistics: 3D leading-order formula

$\boldsymbol{Y} = (\alpha, \eta^2, J_1, J_2)$

$$\langle n^\star(\nu)\rangle \Rightarrow \sum S \int d\boldsymbol{Y}\, n^\star \hat{\mathscr{D}}\hat{\mathscr{D}}\hat{\mathscr{D}}\mathscr{P}_{\rm G}$$

$$\bar{n}^\star(\nu) = G_{00000} + \frac{\sigma_0}{6}\Big[G_{30000}S^{(0)} + 4\gamma G_{21000}S^{(1)} + 3\gamma^2 G_{12000}S_2^{(2)} + \gamma^3 G_{03000}S_1^{(3)} + 4G_{10100}S^{(1)}$$

$$+\frac{8}{3}\gamma G_{01100}S^{(2)} + 6\gamma^2 G_{10010}\left(S_2^{(2)} - S^{(2)}\right) + 3\gamma^3 G_{01010}\left(3S_2^{(3)} - S_1^{(3)}\right) + \frac{75}{14}\gamma^3 G_{00001}\left(\frac{5}{3}S_1^{(3)} - 3S_2^{(3)}\right)\Big]$$

- where $G \equiv G^\star_{ijklm}(\nu) = \dfrac{1}{(2\pi)^{3/2}}\left(\dfrac{\sigma_2}{\sqrt{3}\sigma_1}\right)^3 X_k \displaystyle\int_{a^\star}^{b^\star} dJ_1\, \mathscr{N}(\nu, J_1)\, H_{i-1,j}(\nu, J_1) f^\star_{lm}(J_1)$

$\mathscr{G}_3^\star$

- Redefining $f^\star_{lm}(J_1)$ extends *T. Matsubara (2020)* from peaks all critical points
- $\sigma_0, \gamma = \sigma_1^2/(\sigma_0\sigma_2), S$ can be defined by poly-spectra

Predicted by cosmological perturbation theory

- $\sigma_j^2 \equiv \displaystyle\int \frac{d^3k}{(2\pi)^3} k^{2j} P(k),\ \ \langle \tilde{f}_{\boldsymbol{k}_1}\tilde{f}_{\boldsymbol{k}_2}\tilde{f}_{\boldsymbol{k}_3}\rangle_{\rm c} \equiv (2\pi)^3 \delta_{\rm D}(\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3) B(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3)$

Numerical implementation: perturbation theory (prediction)

- Initial condition ➡️ N-body sim. (particle evolved by gravity) ➡️ nonlinear structure web

  ≈ Gaussian field Fiducial Quijote Highly non-Gaussian

- Linear power spectrum today: $P_{\mathrm{L}}(k, z=0) = \frac{8\pi^2}{25} \frac{\overbrace{\mathscr{A}_s}^{\Omega_{\mathrm{b}}/\Omega_{\mathrm{m}}}}{\Omega_{\mathrm{m}}^2} T^2(k) \frac{k^{n_s}}{H_0^4 k_{\mathrm{p}}^{n_s-1}}$

  $\boldsymbol{\theta} = (\Omega_{\mathrm{m}}, \Omega_{\mathrm{b}}, h, n_s, \sigma_8)$

  note. $\sigma_8^2 = \int_0^\infty \frac{dk}{2\pi^2} k^2 P_{\mathrm{L}}(k, z=0)\, \tilde{W}^2\left(kR_8\right),$ smoothing scale at $8\,h^{-1}\mathrm{Mpc}$

- Standard PT: $B_{3\mathrm{D}}(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) = 2\tilde{W}(k_1R)\tilde{W}(k_2R)\tilde{W}(k_3R)\left[F_2(\boldsymbol{k}_1, \boldsymbol{k}_2) P_{\mathrm{L}}(k_1) P_{\mathrm{L}}(k_2) + \text{cyc.}\right],$

  where $F_2(\boldsymbol{k}_1, \boldsymbol{k}_2) = \frac{5}{7} + \frac{1}{2}\frac{\boldsymbol{k}_1 \cdot \boldsymbol{k}_2}{k_1 k_2}\left(\frac{k_1}{k_2} + \frac{k_2}{k_1}\right) + \frac{2}{7}\left(\frac{\boldsymbol{k}_1 \cdot \boldsymbol{k}_2}{k_1 k_2}\right)^2,$ $\tilde{W}(kR)$: window function

Numerical implementation: finding algorithm (measurement)

Implemented with the assistance of AI

for each grid cell centered at $\boldsymbol{x}_c$ **do**

Compute $\boldsymbol{g} = \nabla f(\boldsymbol{x}_c)$ and $H = \nabla\nabla f(\boldsymbol{x}_c)$ by finite differentiation

Solve Newton offset $\Delta \boldsymbol{x}$: $\partial f / \partial \boldsymbol{x} \simeq \boldsymbol{g} + H \Delta \boldsymbol{x} = 0$

if $|\Delta x_i| < \Delta_i^{\mathrm{grid}}/2$ for all directions **then**

Refine position: $\boldsymbol{x}_\star = \boldsymbol{x}_c + \Delta x$, field value: $f_\star \simeq f(\boldsymbol{x}_c) + \boldsymbol{g} \cdot \Delta \boldsymbol{x} + \dfrac{1}{2} \Delta \boldsymbol{x}^T H \Delta \boldsymbol{x}$

Compute eigenvalues of $H(x_\star)$, remove duplicates

Identified points in selected slices from a 3D field

Numerical implementation: prediction v.s. measurement

$\delta_m(\boldsymbol{x})$: nonlinear matter field

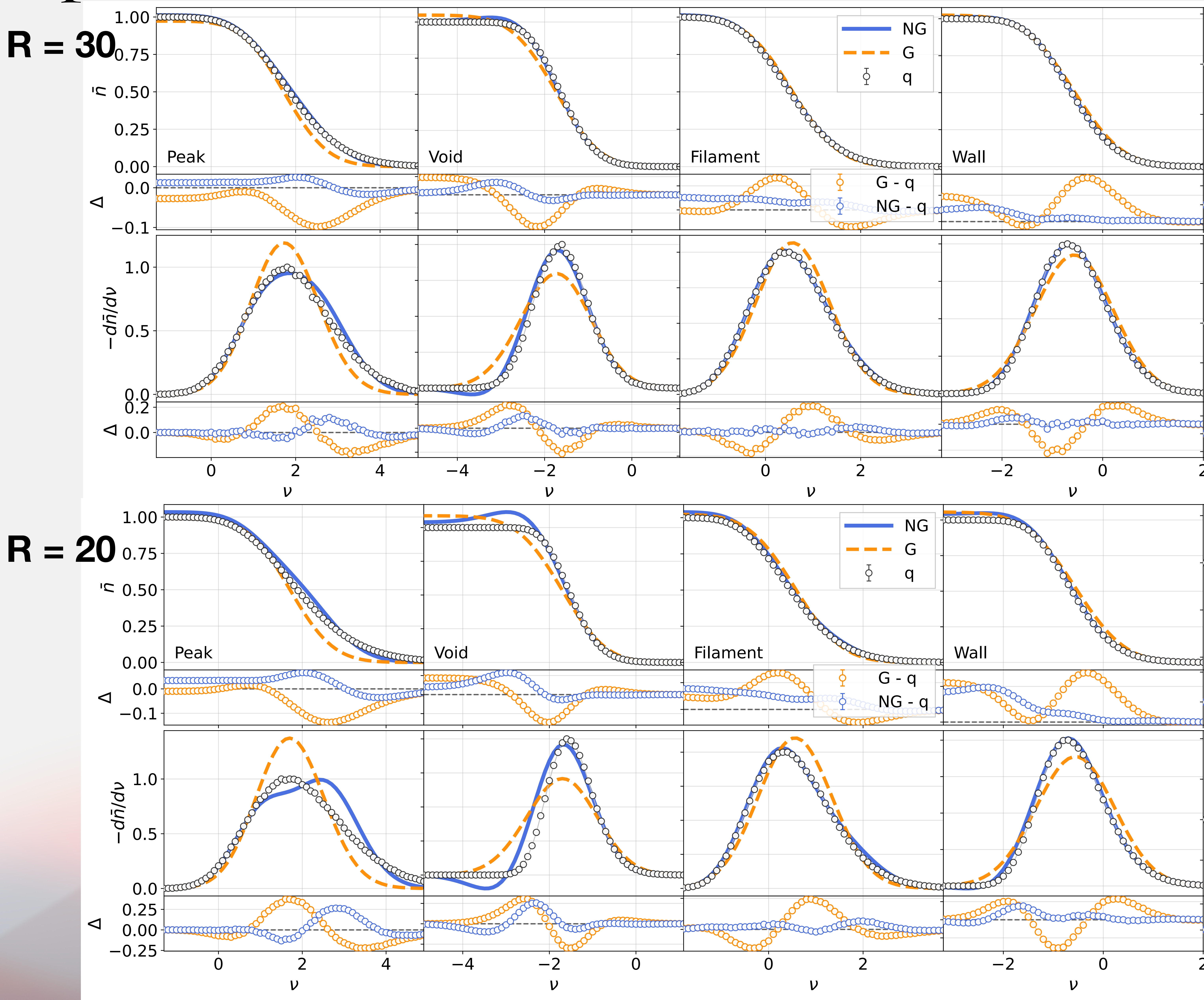
$\tilde{W}(kR) \equiv \exp(-k^2R^2/2)$

$$\delta_m(R,\boldsymbol{x}) = \int d^3\boldsymbol{x}'\, W(|\boldsymbol{x}-\boldsymbol{x}'|/R)\delta_m(\boldsymbol{x}')$$

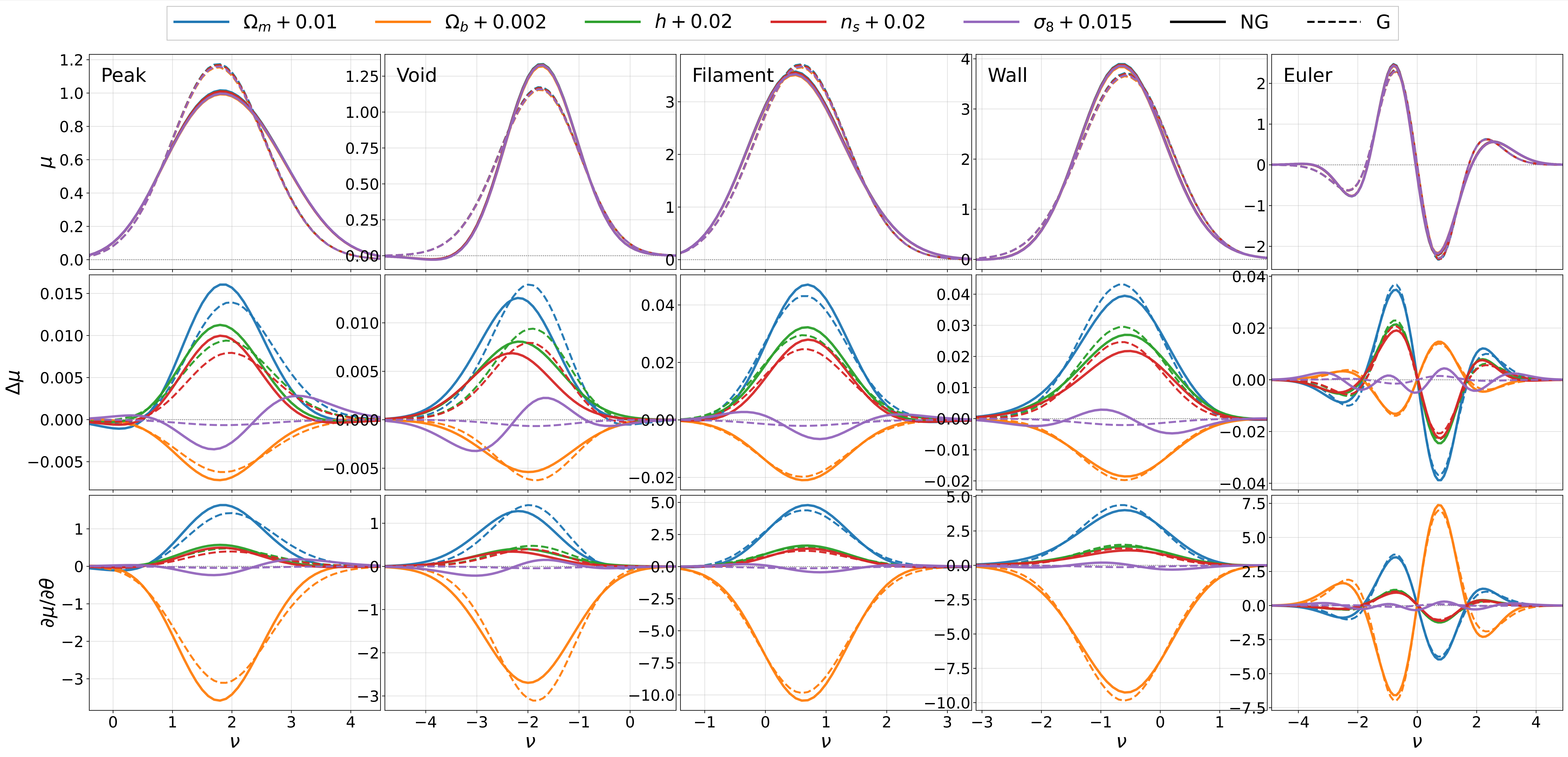
$N_{\rm grid} = 512,\ N_{\rm realization} = 400$

$L_{\rm box} = 1000\, h^{-1}{\rm Mpc}$

ν : threshold
$\bar{n}$: cumulative
$-d\bar{n}/d\nu$: differential
$\mathrm{NG, G}$: theory prediction
q : numerical finder
$\Delta = \text{theory} - q$



Parameter inference: $(\Omega_{\rm m}, \Omega_{\rm b}, h, n_s, \sigma_8)$ sensitivity of $-d\bar{n}^\star/d\nu$ formula



$$\mu^\star(\nu,\theta) \equiv -\frac{d\bar{n}^\star}{d\nu}(\nu,\theta), \quad \Delta\mu^\star = \mu^\star(\theta) - \mu^\star(\theta_{\rm fid}), \quad \frac{\partial\mu^\star}{\partial\theta}(\theta_{\rm fid}) \approx \frac{\mu^\star(\theta_{\rm fid}+\Delta\theta) - \mu^\star(\theta_{\rm fid}-\Delta\theta)}{2\Delta\theta}$$

Parameter inference: Bayesian pipeline

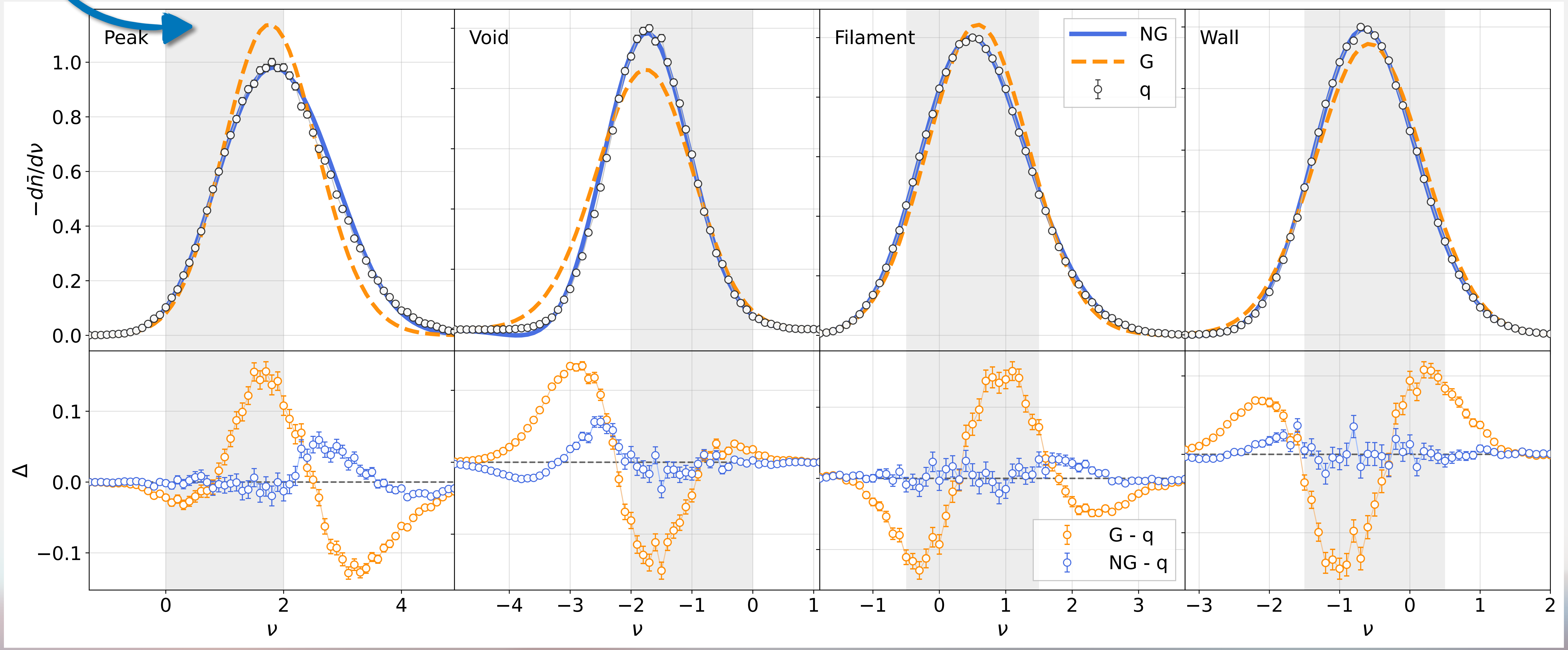
- Data: $\boldsymbol{d} \equiv \left[\bar{q}^{\mathrm{peak}}, \bar{q}^{\mathrm{filament}}, \bar{q}^{\mathrm{wall}}, \bar{q}^{\mathrm{void}}\right]$, Covariance: $\hat{C} = \frac{1}{N_{\mathrm{sim}} - 1} \sum_{s=1}^{N_{\mathrm{sim}}} (\boldsymbol{d}_s - \bar{\boldsymbol{d}})(\boldsymbol{d}_s - \bar{\boldsymbol{d}})^T$

$-d\bar{n}/d\nu$

1000

- Likelihood: $-2\ln\mathscr{L}(\theta) = (\bar{\boldsymbol{d}} - \mu(\theta))\hat{C}^{-1}(\bar{\boldsymbol{d}} - \mu(\theta))^T$, Posterior: $P(\theta\,|\,\bar{\boldsymbol{d}}) \propto \mathscr{L}(\bar{\boldsymbol{d}}\,|\,\theta)P_{\mathrm{prior}}(\theta)$

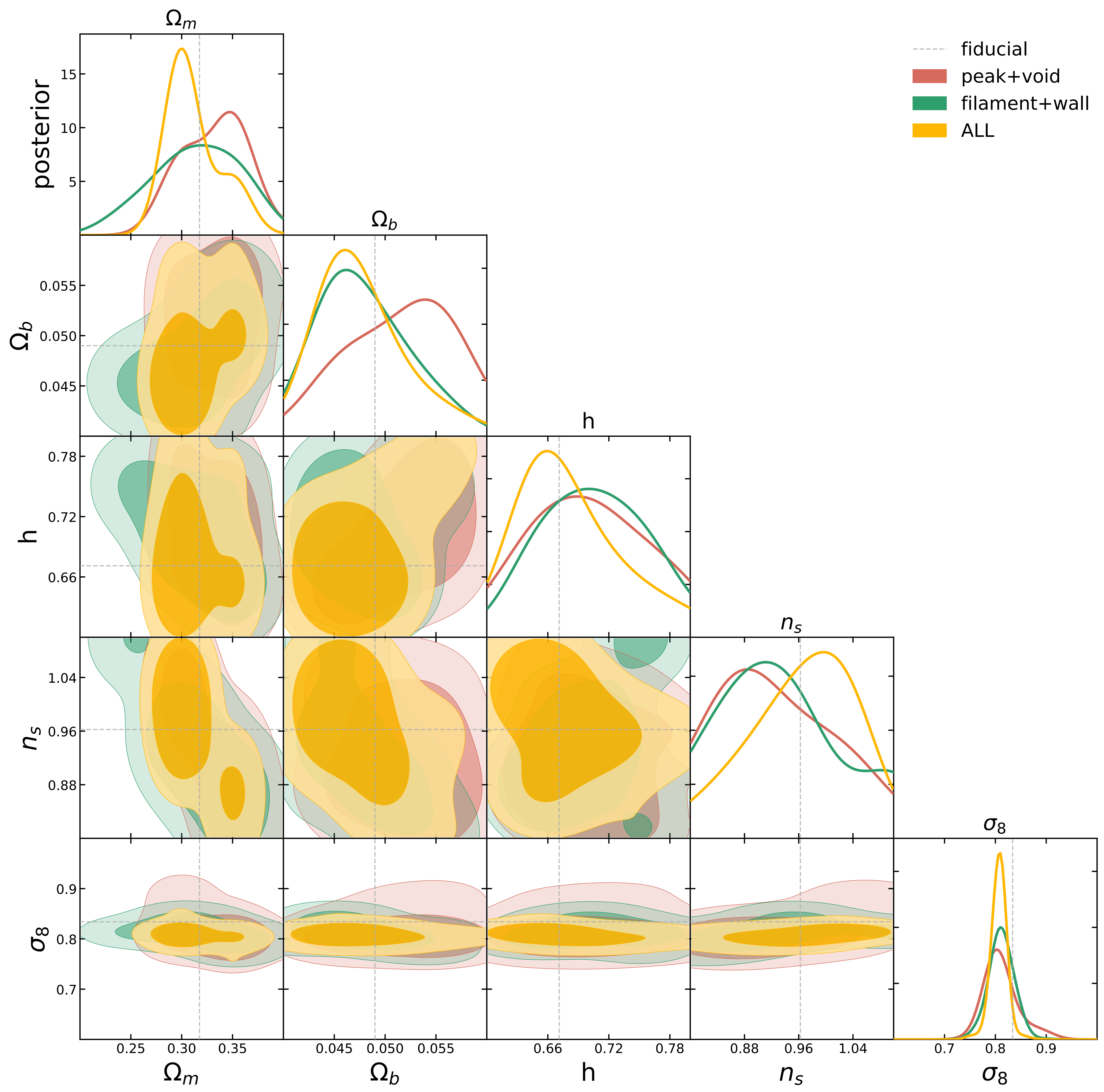
R = 40



Parameter inference

- Power spectrum: CAMB halo model
- Bispectrum: SPT tree-level
- Field: Fiducial Quijote simulation
- Nested sampling method
- $R = 40\,h^{-1}\mathrm{Mpc}, N_{\mathrm{sim}} = 1000$
- ALL = peak+void+fila+wall (44 bins)

Reduce degeneracy



Conclusion

- Peak number density formula in *T. Matsubara (2020)* is extended for all critical points in weak non-Gaussian density fields.

- Leading-order non-Gaussian correction is verified to work much better than the Gaussian prediction at large smoothing scales, e.g. $R = 40\,h^{-1}\mathrm{Mpc}$.

- We found that the leading-order predictions for peaks and voids become less accurate when $R = 20\,h^{-1}\mathrm{Mpc}$, while filaments and walls remain stable.

- Restricting the analysis to the validated regime, the combined critical-point data vector constraints $(\Omega_\mathrm{m}, \Omega_\mathrm{b}, h, n_s, \sigma_8)$ and helps reduce parameter degeneracies beyond Gaussian prediction.

2D Leading-order formula

- Ensemble averaged number density:

$$\bar{n}^{\star}(\nu) = G_{0000} + \frac{\sigma_0}{6}\Big[G_{3000}S^{(0)} + 4\gamma G_{2100}S^{(1)} + 3\gamma^2 G_{1200}S_2^{(2)} + \gamma^3 G_{0300}S_1^{(3)} + 4G_{1010}S^{(1)}$$
$$+2\gamma G_{0110}S^{(2)} + 6\gamma^2 G_{1001}\left(S_2^{(2)} - S^{(2)}\right) + 6\gamma^3 G_{0101}\left(2S_2^{(3)} - S_1^{(3)}\right)\Big]$$

- where $G_{ijkl}(\nu) = \frac{1}{4\pi}\left(\frac{\sigma_2}{\sigma_1}\right)^2 X_k \int_{a^\star}^{b^\star} dJ_1\, \mathcal{N}(\nu, J_1)\, H_{i-1,j}(\nu, J_1) f_l^\star(J_1)$

Result: 2D toy model field

$N_{\mathrm{grid}} = 2048,\ N_{\mathrm{realization}} = 100$

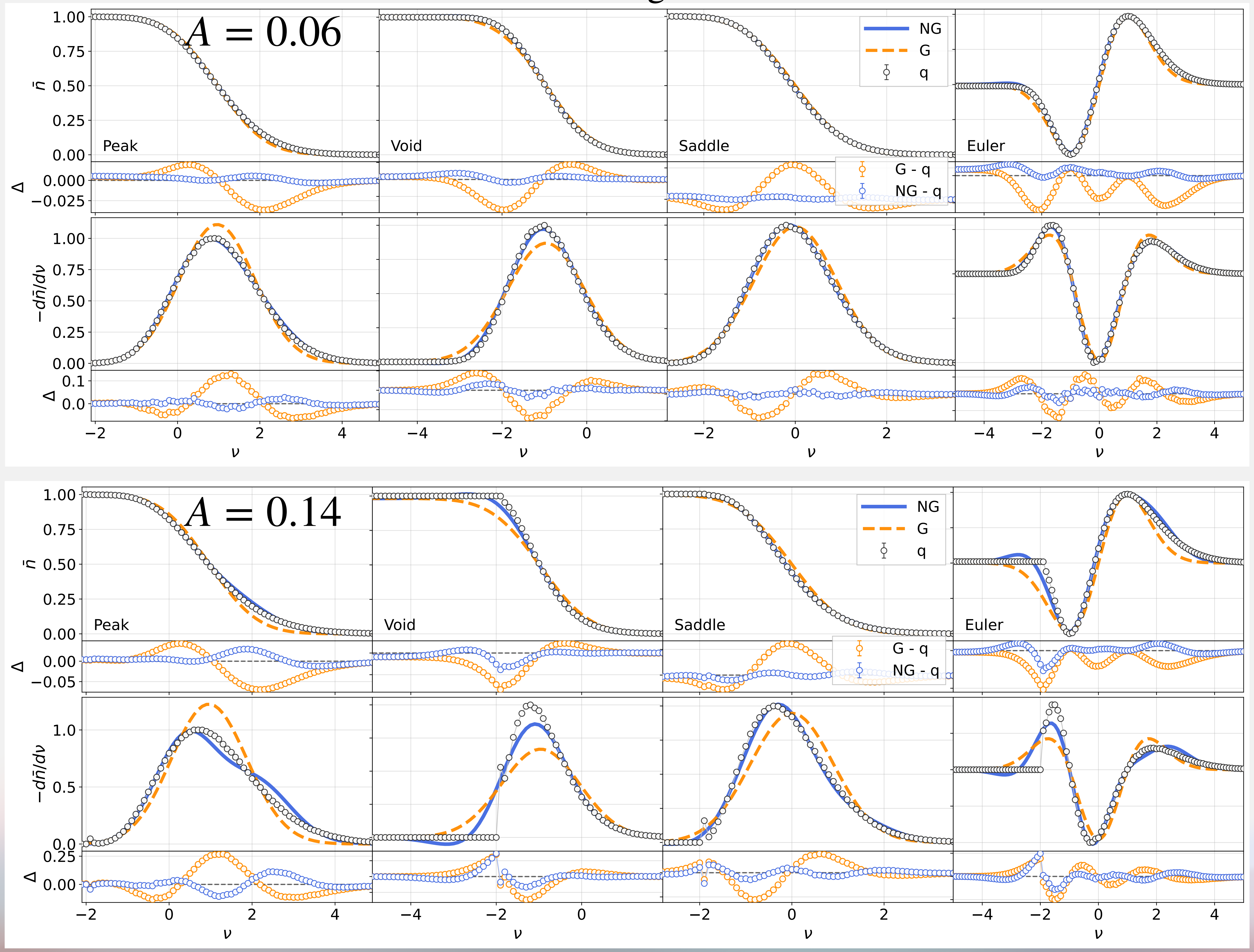
$P_{f_g f_g}(k)$

↓ Fourier transform

$f = f_g + A f_g^2$ → $B_{fff}(k_1, k_2, \mu)$

$\sim \mathcal{N}(0,1)$

Wick's theorem



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